

Research App — Rich Feature Architecture & Implementation Guide

A Comprehensive Technical Specification of All Implemented Features, Native Document Viewer, SQLite Engine, and Ecosystem Interoperability Contracts

Core Product Principle: The Research App is the persistent evidence, discovery, verification, and synthesis engine of the personal software ecosystem.
Its defining identity is: **Sources** → **Excerpts** → **Evidence** → **Claims** → **Concepts** → **Synthesis** → **Outputs**, with provenance preserved throughout.
Architectural Boundary: Stylus handwriting and ink drawing belong exclusively to *Stylus Notes*. The Research App embeds local-first PDF and Markdown viewing purely for reading and extracting evidence coordinates directly into propositions.

1. Ecosystem Role & Canonical Data Boundaries

The personal ecosystem operates under the principle of *independent applications sharing a common language*. Each application maintains strict canonical ownership over its specific domain primitives, communicating via contracts and deep links.

Application	Core Purpose	Canonical Domain Ownership	Research App Integration Bridge
Research	Discover, Verify & Synthesize	Projects, Questions, Sources, Claims, Evidence, Matrices, Syntheses	Core Engine; exposes URI scheme research://...
Likhoji	Typing & Keyboard IME	IME Keyboards, Tries, Clipboard, Dictionaries	Sends text selections via com.ecosystem.action.SAVE_RESEARCH
Boloji	Voice Dictation Companion	Speech Recognition Streams, Audio Buffers	Fires voice idea broadcast ACTION_VOICE_IDEA
Stylus Notes	Capture, Handwriting & Notes	Ink strokes, Canvas, Note notebooks, Drawing	Deep link handoff: stylus://open?sourceId={id}&page;={n}
Productivity	Plan, Schedule & Execute	Tasks, Deadlines, Time Blocks, Reminders	Research requests tasks via ACTION_CREATE_TASK
DegreeTrack	Curriculum & Learning Mastery	Courses, Modules, Syllabi, Mastery scores	Links research concepts to academic curriculum topics
Health	Physiological & Health State	Vitals, Labs, Symptoms, Medications	Strict Isolation: Zero automatic access or background sync

2. Detailed Specification of Implemented Features

Below is the complete, granular breakdown of all 23 features, components, and architectural subsystems built into the application codebase.

Feature 1: Research Command Center (Dashboard)

The top-level operational mission control of the app surfacing actionable project metrics and immediate document access.

- **Metric Cards:** Real-time counters showing Active Projects and Pending Inbox count with 1-tap navigation.
- **Pipeline Orientation:** Visual guide indicating the active phase: DISCOVER → VERIFY → CONNECT → SYNTHESIZE.
- **Project Card Grid:** Shows project titles, primary research questions, tags, and status chips (ACTIVE, IN_REVIEW, SYNTHESIZED).
- **Quick Creation & Open Document:** Header actions allowing immediate creation of research initiatives or direct file viewing.

Feature 2: Research Project Workspace

The primary container for scientific investigation, replacing unstructured loose PDF directories.

- **Container Isolation:** Every project encapsulates its own literature base, claims, evidence links, and custom matrices.
- **Header Overview:** Displays source count, primary question, and immediate shortcut actions to the Evidence Board and Matrix.
- **Manual Source Ingestion:** Modal dialog to record non-indexed sources, offline documents, or manual citations.
- **Fast Context Switching:** Seamlessly navigate between literature queue, claim extraction, and comparative matrix.

Feature 3: Structured Research Question Builder (PICO Framework)

Guides broad scientific curiosity into a rigorous, searchable, and verifiable research question.

- **PICO Fields:** Dedicated inputs for Population/Patient (P), Intervention/Exposure (I), Comparison/Control (C), and Outcome (O).
- **Search Query Derivation:** Programmatically converts PICO components into boolean search terms for PubMed and arXiv.
- **Question Versioning:** Tracks revisions as empirical findings refute initial hypotheses.
- **Scientific Guardrail:** Ensures that discovery queries match clinical and technical research standards from the outset.

Feature 4: 8-State Reading Queue Management

A rigorous literature lifecycle engine to track reading workflows, screen studies, and prioritize critical papers.

- **Granular States:** INBOX (unprocessed), TO_SCREEN, TO_READ, READING, READ, KEY_PAPER, ARCHIVED, and EXCLUDED.
- **Scrollable Tab Filtering:** Instant one-tap filtering in the workspace to view only papers awaiting screening or critical benchmarks.
- **State Transition Dropdown:** Contextual menu on every paper card enabling rapid state progression.
- **Screening Audit Trail:** Papers marked EXCLUDED retain their exclusion rationale for systematic reviews.

Feature 5: Structured Paper Cards & Metadata System

Papers are modeled as structured knowledge entities rather than inert binary files.

- **Comprehensive Metadata:** Stores Title, Authors list (JSON), Journal, Publication Year, Abstract, DOI, PMID, arXiv ID, and URLs.
- **External ID Anchoring:** Primary identification uses canonical keys (DOI, PMID, arXiv ID).
- **Inline Abstract Preview:** 3-line truncated abstract with visual hierarchy for rapid scanning.
- **Reading & Provenance Badges:** Visual color-coded tags indicating both current reading status and metadata origin.

Feature 6: Claim-Evidence Proposition System

The core distinguishing pillar of the Research App: knowledge is organized around claims rather than documents.

- **First-Class Claim Entity:** Claims represent testable scientific assertions (e.g., 'Model generalizability degrades on non-hospital cohorts').
- **Lifecycle Statuses:** PROPOSED (unverified hypothesis), SUPPORTED, CONTESTED (conflicting findings), REFUTED, and SYNTHESIZED.
- **Multi-Study Aggregation:** A single claim links evidence excerpts across 5, 10, or 50 independent papers.
- **Research Notes:** Dedicated notes field for researcher commentary and qualitative synthesis.

Feature 7: Evidence Cards with Granular Coordinate Provenance

Extracted observations become Evidence Cards that preserve backward-traceable pointers to original sources.

- **Coordinate Pointers:** Captures exact Page Number, Character Offset Start/End, Section header, and bounding box.
- **Excerpt Preservation:** Stores verbatim quotes from the original publication alongside the researcher's interpretation.
- **'Open Original' Action:** One-tap coordinate trigger jumping directly to the exact page in the source material.
- **Audit Transparency:** Guarantees that no fact or summary exists detached from its empirical origin.

Feature 8: Multi-Relationship Evidence Linking

Encodes the exact relationship between empirical evidence and a scientific proposition.

- **Relationship Taxonomies:** SUPPORTS, CONTRADICTS, MIXED, BACKGROUND, RELATED, and UNDETERMINED.
- **Color-Coded Badging:** Green for SUPPORTS, Red for CONTRADICTS, Amber for MIXED, Neutral Gray for BACKGROUND.
- **Relationship Review Notes:** Allows documenting why an excerpt contradicts (e.g., 'Subgroup analysis revealed failure in pediatric cohort').
- **Extensible Graph Edges:** Relational edge table mapping allows many-to-many Claim ↔ Evidence relationships.

Feature 9: Automated Contradiction Finder

Automatically flags scientific disagreements across literature connected to the same proposition.

- **Conflict Detection Algorithm:** Scans linked evidence and alerts whenever both SUPPORTS and CONTRADICTS edges exist on a single claim.
- **Visual Warning Banner:** Displays an immediate prominent red warning header on the Claim Card alerting to opposing findings.
- **Side-by-Side Comparison:** Groups opposing evidence cards beneath the proposition for methodology evaluation.
- **AI Candidate Generation:** Offline and BYOK AI engines generate candidate contradiction alerts for human review.

Feature 10: Multi-Study Evidence Matrix with Cell-Level Provenance

A structured comparative grid for multi-paper synthesis, essential for clinical trials, systematic reviews, and ML evaluations.

- **Configurable Schema:** Default columns for Population/N, Study Design, Intervention/Method, Comparator, Primary Outcome, Results, and Limitations.
- **Horizontal Scroll Grid:** Smooth, low-latency table rendering capable of viewing dozens of studies side-by-side.
- **Cell-Level Provenance Modal:** Tapping any table cell opens a modal showing its exact source publication and extraction details.
- **Persistent Cell Edits:** Cell edits are saved directly to SQLite with verification state tags.

Feature 11: Universal Research Inbox

A frictionless capture hub eliminating organizational friction when browsing or reading.

- **Multi-Modal Capture:** Accepts Raw Text, DOIs, PMIDs, Web URLs, Text Snippets, PDF Paths, and Voice Dictations.
- **Source App Attribution:** Automatically stamps incoming captures with their originating app ('boloji', 'likhoji', 'manual').
- **Triage to Project:** One-tap project assignment converts an inbox snippet into a structured Source or Question.
- **Manual Quick Capture FAB:** Allows typing or pasting notes, DOIs, or queries with instant category tagging.

Feature 12: Live External Literature Discovery (PubMed & arXiv)

Unified literature search directly inside the app without switching to external browser tabs.

- **PubMed Integration:** Queries NCBI E-utilities returning titles, authors, journals, years, DOIs, and PMIDs.
- **arXiv Integration:** Queries arXiv Export Atom XML API returning preprints, abstracts, arXiv IDs, and publication timestamps.
- **1-Tap Project Import:** Adds discovered papers directly into the active project workspace reading queue.
- **Network Resiliency:** Built with OkHttp timeouts, UTF-8 URL encoding, and background Dispatchers.IO threading.

Feature 13: Identity & De-duplication Registry

Prevents duplicate records when papers are imported from multiple sources or preprints.

- **DOI De-duplication:** Checks SQLite indices before inserting; returns a clear warning if DOI already exists.
- **PMID De-duplication:** Ensures PubMed entries are not imported twice into the research library.
- **Canonical Key Matching:** Normalizes trimmed, lowercased identifiers across Crossref and PubMed records.
- **Library Integrity:** Protects against database bloat and fragmented evidence connections.

Feature 14: Provenance Everywhere & Verification States

Guarantees that AI-generated inferences never silently masquerade as confirmed scientific truth.

- **Origin Types:** SOURCE_METADATA, DIRECT_EXCERPT, USER_WRITTEN, AI_EXTRACTED, and AI_INFERRED.
- **Verification States:** UNVERIFIED, USER_REVIEWED, CONFIRMED, and FLAGGED_CONTRADICTIONARY.
- **Attribution Packet:** Records createdBy, timestamp, modelName (e.g. 'gemini-1.5-flash'), and promptFingerprint.
- **UI Badging:** Unverified AI extractions are visibly badged in purple; user-confirmed facts are badged in green.

Feature 15: BYOK AI Layer & Heuristic Offline Engine

A Bring-Your-Own-Key AI architecture that ensures core research functionality works 100% offline.

- **Provider Interface:** Clean abstract contract defining extractStructuredFields, detectContradictions, and synthesizeGroundedReview.
- **Heuristic Offline Engine:** Built-in offline fallback extracts study design (RCT, Cohort, Case-Control) and sample sizes ($n = \dots$) via regex.
- **Offline Contradiction Logic:** Compares evidence edges without internet connectivity.
- **Context Scoping:** AI prompts receive only user-selected excerpts and papers, never broadcasting the entire research library.

Feature 16: Stylus Notes Deep Integration Bridge

Delegates rich PDF inking, pen drawing, and handwriting annotations to Stylus Notes.

- **Deep Link Launch:** Fires Intent with URI `stylus://open?sourceId={id}&page={n}` to open the exact document and page.
- **Preserved Boundaries:** Research App avoids duplicating heavyweight stylus and drawing engines.
- **Coordinate Return:** Stylus Notes returns highlight coordinates, colors, and text snippets back to Research.
- **Source Traceability:** Every highlight in Stylus Notes remains linked to its canonical Source in Research.

Feature 17: Productivity Inter-App Task Dispatcher

Transforms research findings into actionable execution without becoming another task manager.

- **Task Request Intent:** Dispatches `com.ecosystem.productivity.ACTION_CREATE_TASK` with task title, notes, and deep link.
- **Standard Action Types:** 'Read Paper', 'Extract Methods', 'Verify Citation', and 'Review Evidence Matrix'.
- **Backward Reference:** Productivity manages execution; Research retains a reference to the resulting task.
- **Workflow Automation:** Moving a paper to 'TO_READ' can automatically suggest creating a reading task in Productivity.

Feature 18: Likhoji IME & Boloji Voice Companion Handlers

Captures thoughts, dictations, and text selections system-wide without opening the Research App.

- **Likhoji Broadcast Handler:** Receives `com.ecosystem.action.SAVE_RESEARCH` and captures selected text directly to the Inbox.
- **Boloji Voice Handler:** Receives `ACTION_VOICE_IDEA` containing voice dictation prompts from the system floating mic.
- **Background Processing:** Incoming intents are parsed and saved to SQLite asynchronously via `lifecycleScope`.
- **Instant Feedback:** Displays native Android Toast confirmations upon successful capture.

Feature 19: Research Contract v1 URI & Deep Link System

Provides stable, platform-wide URIs so any ecosystem app can reference Research objects without copying them.

- **URI Schemes:** `research://project/{id}`, `research://source/{id}?page={n}`, `research://claim/{id}`, `research://evidence/{id}`.
- **Standard Parser:** `ResearchContractV1.parseUri` safely decomposes incoming URIs into typed `DeepLinkTarget` sealed classes.
- **Manifest Intent Filters:** `AndroidManifest` registers `scheme='research'` with `VIEW`, `DEFAULT`, and `BROWSABLE` categories.
- **Zero-Copy Linking:** Stylus Notes, DegreeTrack, and Productivity store URIs rather than duplicating heavy paper records.

Feature 20: Local-First Native SQLite Database Engine

Ultra-fast, zero-annotation-processor `SQLiteOpenHelper` database guaranteeing reliability and instant compilation.

- **Direct SQLite Architecture:** Eliminates KSP annotation processor bytecode generation errors and compiler daemon crashes.
- **8 Relational Tables:** `projects`, `sources`, `claims`, `evidence`, `claim_evidence_edges`, `inbox_items`, `evidence_matrices`, `matrix_cells`.
- **Indexed Foreign Keys:** High-performance B-tree indices on `doi`, `pmid`, `project_id`, `reading_status`, and `edge` keys.
- **Reactive StateFlows:** Repository queries execute on `Dispatchers.IO` and emit reactive `StateFlow` updates to Jetpack Compose.

Feature 21: Security, Privacy & Android Keystore Encryption

Protects unpublished scientific hypotheses, personal notes, and API credentials.

- **Local-First Privacy:** All library metadata, claims, excerpts, and matrix data reside strictly in app-sandboxed SQLite storage.
- **Encrypted Secrets:** BYOK API keys are encrypted with AES-256-GCM via Android Keystore and `EncryptedSharedPreferences`.
- **Zero Cloud Telemetry:** No analytics, tracking, or unexpected third-party logging.
- **Health Data Isolation:** Hard boundary preventing automatic querying or importing of user medical records.

Feature 22: GitHub CI/CD Pipeline & Automated Release Delivery

Automated build and delivery pipeline enabling continuous testing and direct APK downloads.

- **Automated GitHub Workflow:** `.github/workflows/build-apk.yml` triggers on pushes, PRs, and tags.
- **Test Automation:** Runs `./gradlew testDebugUnitTest` verifying URI contracts, provenance, and document parsing.
- **APK Artifact Upload:** Compiles `assembleDebug` and uploads `ResearchApp-Debug-APK` artifact (22.4 MB).
- **Automatic Release Publishing:** Pushing tags (e.g. `v1.0.2`) triggers `softprops/action-gh-release@v2` to publish releases.

Feature 23: Native PDF & Markdown Document Viewer with Evidence Extraction

A local-first document reader specifically designed for scientific papers and research notes, strictly omitting stylus tools.

- **Native PDF Engine:** Built-in Android `PdfRenderer` renders pages at high display resolution with smooth zoom and page navigation.
- **Page Navigation & Jump:** Bottom control bar with Previous/Next buttons, zoom controls, and a 'Jump to Page' modal.
- **Rich Markdown Renderer:** Zero-dependency Compose renderer supporting headers (`#`, `##`, `###`), bold/italics, blockquotes, bullet lists, and code blocks.
- **Plaintext & Monospace Viewer:** Clean viewing for `.txt`, `.bib`, `.csv`, and `.json` assets.
- **'Extract Evidence' Action:** Persistent action button pre-filling the current page coordinate and source to capture excerpts into the evidence graph.

3. Local SQLite Database Schema Specification

The relational database schema is implemented in ResearchDatabase.kt (DB version 1, research_ecosystem.db). It provides strong relational integrity, index coverage, and complete field traceability.

Table Name	Primary Key	Key Columns & Indices	Domain Entity
projects	id (UUID TEXT)	title, description, primary_question, status, tags, created_at, updated_at	ResearchProject
sources	id (UUID TEXT)	project_id, title, authors_json, journal, year, doi, pmid, arxiv_id, url, abstract_text, local_pdf_path, reading_status, study_type, priority, rationale, origin_type, verification_state, created_by <i>Indices: project_id, doi, pmid, reading_status</i>	Source
claims	id (UUID TEXT)	project_id, proposition, status, notes, origin_type, verification_state, created_by, created_at, updated_at <i>Index: project_id</i>	Claim
evidence	id (UUID TEXT)	project_id, source_id, page_number, char_offset_start/end, section, bounding_box, excerpt_text, user_interpretation, relationship_type, origin_type, verification_state, created_by <i>Indices: project_id, source_id</i>	Evidence
claim_evidence_edges	(claim_id, evidence_id)	relationship_type, review_note, created_at <i>Composite PK; joins claims and evidence</i>	ClaimEvidenceEdge
inbox_items	id (UUID TEXT)	raw_type, raw_content, source_app, assigned_project_id, processed_source_id, status, note, created_at <i>Index: status</i>	InboxItem
evidence_matrices	id (UUID TEXT)	project_id, title, columns_json, created_at <i>Index: project_id</i>	EvidenceMatrix
matrix_cells	(matrix_id, source_id, column_key)	cell_value, evidence_id, verification_state <i>Composite PK; cell-level provenance link</i>	MatrixCell

4. Automated Verification & Quality Assurance Report

All domain contracts, document rendering helpers, contradiction detection heuristics, and APK assembly were validated through automated test pipelines.

Test Suite / Command	Scope & Assertions	Status
DocumentViewerTest	Validates format detection (.pdf, .md, .txt), markdown inline parsing (**bold**, `code`), and evidence coordinate preservation.	PASSED
ResearchContractV1Test	Validates URI generation and parsing for projects, sources, claims, and evidence. Ensures scheme prefix research:// integrity.	PASSED
ProvenanceAndAiTest (Flags)	Validates that AI extractions are flagged isAiDerived=true and blocked from isConfirmed=true until reviewed.	PASSED
ProvenanceAndAiTest (Contradiction)	Feeds opposing SUPPORTS and CONTRADICTS evidence pairs for a multimodal AI claim; verifies automated candidate conflict identification.	PASSED
ProvenanceAndAiTest (Extraction)	Feeds unstructured clinical text; verifies regex heuristic extraction of study design (RCT) and sample size (n = 1,420).	PASSED
assembleDebug	Full compilation, resource packaging, dexing, and signing. Generated standalone APK: app-debug.apk (22.4 MB).	PASSED
GitHub CI/CD Pipeline	Automated GitHub Actions runner on Ubuntu executes Gradle unit tests and APK packaging on every commit and release tag.	PASSED

5. Future Phasing & Architectural Dependency Map

With the core foundation, reading queue, and native document viewer established, subsequent phases will expand into citation management, live LLM integration, and knowledge graphs.

Phase	Focus & Modules	Key Deliverables & Dependencies	Status
Phase 0 & 1	Core Foundation, DB, UI & Discovery	Workspace, PICO, Reading Queue, Evidence Board, Matrix, Inbox, PubMed/arXiv Discovery, Local SQLite, Android APK	COMPLETE
Phase 2 (Rev)	Native PDF & Markdown Document Viewer	Native PdfRenderer, rich Markdown Composable, monospace text viewer, coordinate evidence extraction (no stylus tools)	COMPLETE
Phase 3	Source-Grounded BYOK AI Pipeline	Direct Gemini 1.5 Flash / Claude Haiku / OpenAI integration with explicit paper selection scopes	PLANNED
Phase 4	Citation Management & Lit Review	BibTeX / RIS / CSL-JSON citation exporter, structured literature review outline builder	PLANNED
Phase 5	Cross-App Event Dispatcher	Automated Productivity task generation from reading states, DegreeTrack curriculum link resolver	PLANNED
Phase 6	Knowledge Graph & Concept Pages	Recursive SQLite Common Table Expression (CTE) graph queries across multi-project concepts	PLANNED
Phase 7+	Specialized Clinical & AI Modes	PRISMA systematic review screening flow, Medical AI evaluation templates (AUROC, splits, baselines)	FUTURE